

Gemma 4 12B

256

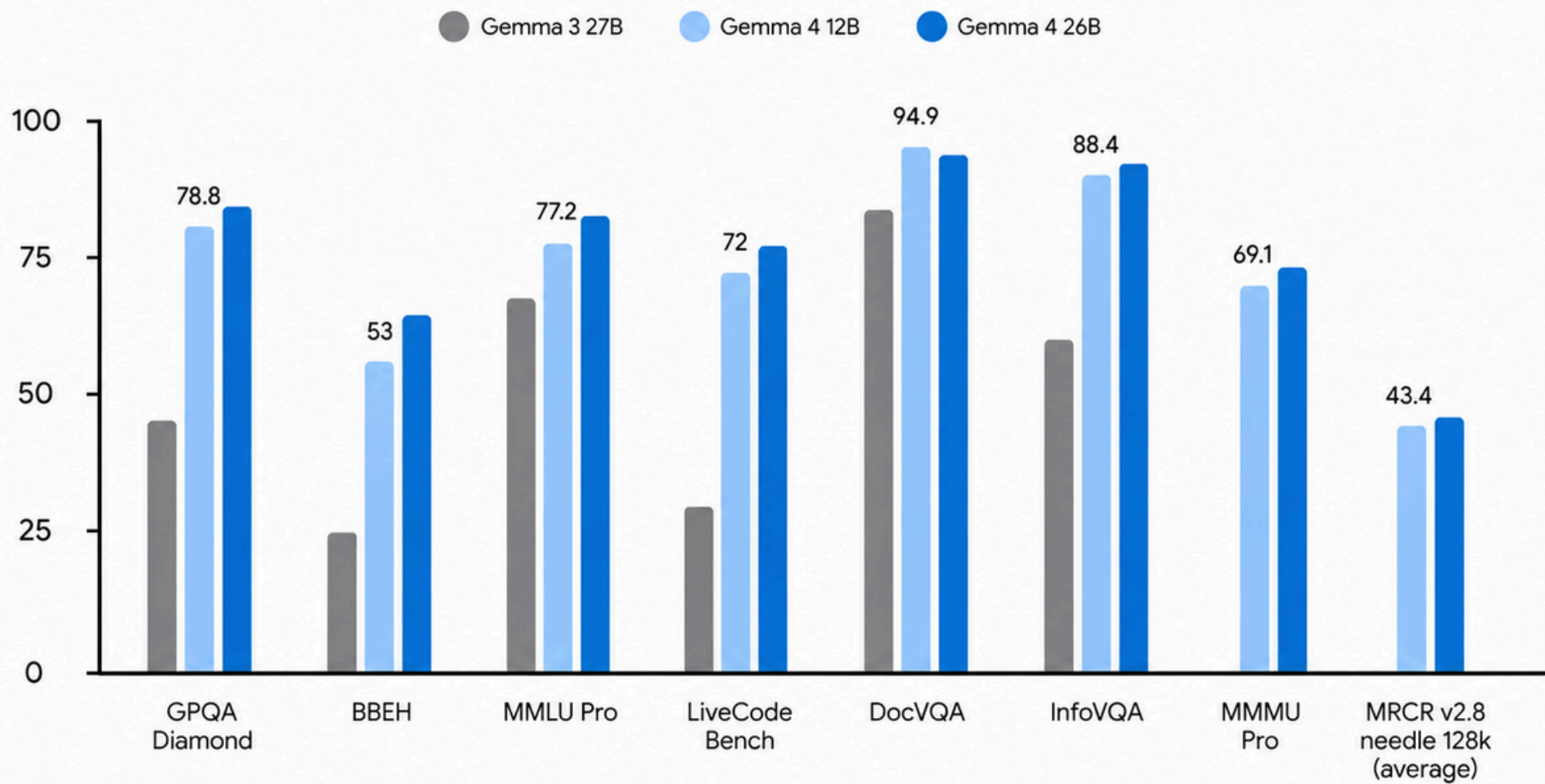
Context
Window

35+

Languages

4

Modalities



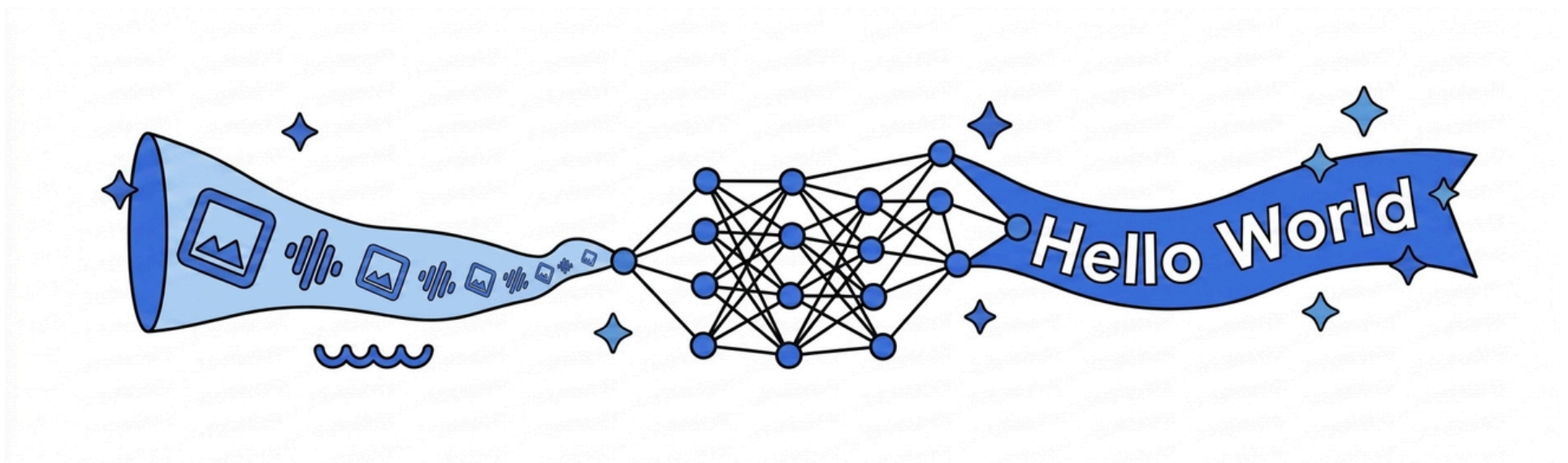
What is Gemma 4 12 B?

Gemma 4 12B Unified is Google's mid-sized open-source multimodal model; built to run natively on your laptop. It understands text, images, audio & video in one model.

Key Features

Everything packed in one open-source model:

-  Text Generation & Long-context Reasoning
-  Code Completion, Correction & Function Calling
-  Vision Understanding - Images & Documents
-  Audio: Speech Recognition + Translation
-  Video Understanding (processed as frames)
-  35+ Languages out-of-the-box, 140+ pre-training
-  Agentic Workflows with Tool/Function Use



Why Google Built it?



The Problem

- E2B/E4B: limited reasoning
- 26B A4B/31B: need servers
- No middle ground for laptop developers!



The Solution

- 12B: strong reasoning + multimodal on a laptop
- Best of both worlds for local deployment

Gemma 4 Release Timeline:



Mar 31, 2026

E2B, E4B, 31B, 26B A4B launched



Apr 16, 2026

MTP Drafters released



Jun 3, 2026

12B Unified laptop-first!

Architecture Deep Dive



Vision Processing

- 35M parameter lightweight vision embedder
- Raw 48×48 pixel patches → single matrix multiply
- Spatial info via factorized coordinate lookups
- No heavy multi-layer encoder overhead



Audio Processing

- No conformer-based audio encoder
- Slices raw 16kHz audio into 40ms frames
- Linearly projects frames into LLM input space
- Supports ASR, translation & diarization



Hybrid Attention

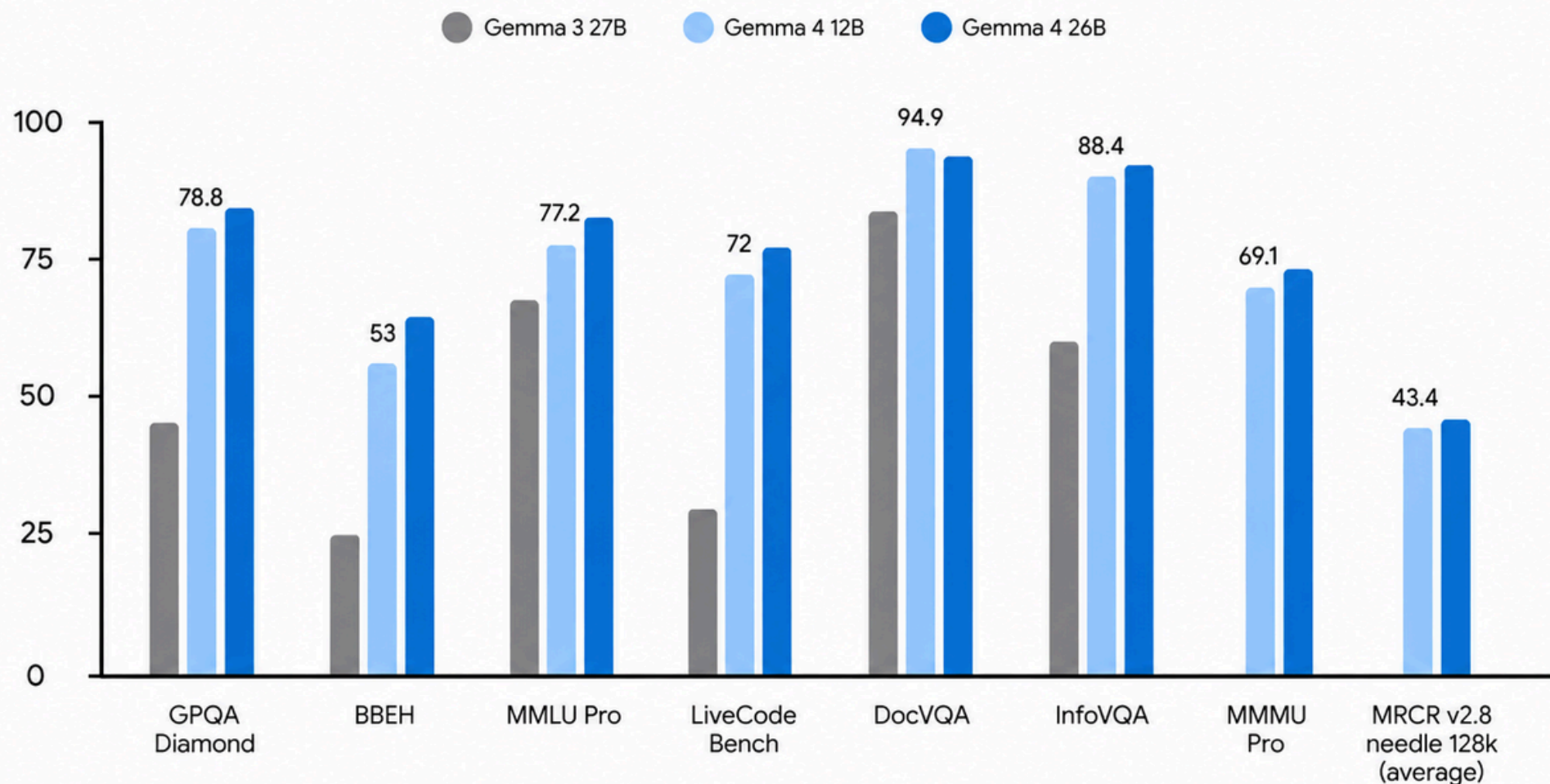
- 1024-token sliding window (local attention)
- Interleaved with full global attention layers
- Final layer is always global attention
- Proportional RoPE for 256K long-context



MTP Drafters

- Multi-Token Prediction for faster decoding
- Small draft model predicts future tokens
- Target model verifies in parallel
- Faster inference, same output quality

Benchmarks & Comparisons



★ Key Insight: 12B sits firmly between E4B and 26B A4B; punching above its weight class!

How to Access & Run

Multiple ways to get Gemma 4 12B running today:



```
# Step 1: Install Ollama from  
ollama.com/download
```

```
# Step 2: Pull & run the model  
ollama run gemma4:12b
```

```
# Step 3: Use with Python SDK  
import ollama  
response = ollama.chat(model='gemma4:12b',  
    messages=  
[{'role': 'user', 'content': 'Hello!'}])
```



kaggle vLLM



Ollama



LM Studio

